



MISO DPP 2019 Cycle2 Michigan Area Study Phase 2 Report

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(Final)
Rev.5

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1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2019 Cycle Michigan Area Phase 2 (Michigan Area DPP-2019-Cycle Phase 2). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies:

- ATC
- MPower
- Consumers Energy
- PJM
- NIPSCO
- DTE
- Wolverine
- ITC

1.1. Project List

The original interconnection requests for DPP 2019 Cycle Phase 1 Michigan Area had a total of 34 generation projects: J1192, J1194, J1196, J1203, J1206, J1207, J1210, J1211, J1224, J1226, J1248, J1255, J1262, J1292, J1297, J1307, J1310, J1319, J1320, J1322, J1327, J1329, J1331, J1350, J1370, J1375, J1379, J1389, J1394, J1399, J1401, J1403, J1430, J1449 and J1472. Therefore, there were total 38 projects generation projects with a combined nameplate rating of 4190.7 MW (ERIS) & 4165.7 MW (NRIS) in Phase1. Projects withdrawn at Decision point 1 are J1206, J1211, J1322, J1327, J1394, J1449. At Decision point 1, J1207 reduced the NRIS from 50 MW to 49.5 MW and J1255 reduced from 200 MW to 100 MW. There are total 28 projects generation projects with a combined nameplate rating of 3520.7 MW (ERIS) & 3495.7 MW (NRIS) in Phase2. The detailed list of Michigan Area DPP Phase 2 is shown below in Table 1, and the Phase 2 study was kicked off on February 16, 2021.

Table 1: List of DPP 2019 Cycle 1 Michigan Area Phase 1 Projects

Project #	Fuel	Summer MW	Transmission Owner	POI Name	County	State	Service Type
J1192	Solar	35	METC	Oneida 138kV Substation	Eaton County	MI	NRIS
J1194	Wind	225.2	METC	Murphy 138 kV Substation	Midland County, Saginaw County	MI	NRIS

Project #	Fuel	Summer MW	Transmission Owner	POI Name	County	State	Service Type
J1196	Solar	200	ITC	Greenwood - Rapson (Banner) 345 kV Line Tap	Sanilac County	MI	NRIS
J1203	Solar	125	METC	Thetford 345kV Substation	Genesee County	MI	NRIS
J1207	Gas	50	METC	Newton 138kV Substation	Branch County	MI	NRIS
J1210	Solar	50	METC	Whittemore - Twining 138kV Line	Iosco County	MI	NRIS
J1224	Solar	170	ITC	Majestic - Milan 345 kV Line Tap	Washtenaw County	MI	NRIS
J1226	Solar	200	METC	Morocco 345 kV Substation	Lenawee County	MI	NRIS
J1248	Solar	100	METC	Marshall - Black Stone 138 kV line	Calhoun County	MI	NRIS
J1255	Solar	100	METC	Marquette - Vergennes 138 kV Line Tap	Ionia County	MI	NRIS
J1262	Gas	25	ITC	Waterman 120kV Substation	Wayne County	MI	ERIS
J1292	Wind	198	METC	Thompson Road 138kV Substation	Ionia County	MI	NRIS
J1297	Solar	170	METC	Cornell - Bell Road 138 kV Line	Shiawassee County	MI	NRIS
J1307	Solar	150	METC	Oneida - Battle Creek 345kV Line	Eaton County	MI	NRIS
J1310	Solar	125	METC	Blackstone - Marshall 138 kV Line	Jackson County	MI	NRIS
J1319	Wind	117.5	ITC	Rapson 120kV Substation	Huron County	MI	NRIS
J1320	Solar	150	METC	Coldwater - Batavia 138kV Line	Branch County	MI	NRIS
J1329	Battery Storage	20	ITC	Majestic - Milan 345 kV Line Tap	Washtenaw County	MI	NRIS
J1331	Solar	100	ITC	Greenwood 120 kV Substation	St. Clair County	MI	NRIS
J1350	Solar	150	METC	Milan - Monroe - Morocco 345 kV Point	Monroe County	MI	NRIS
J1375	Solar	100	METC	Du Pont - Cobb -White Lake 138kv	Muskegon County	MI	NRIS
J1379	Solar	200	METC	Eureka - Vestaburg 138 kV Line Tap	Montcalm County	MI	NRIS
J1389	Solar	120	METC	Murphy Station - Nelson Road 345 kV Line tap	Saginaw County	MI	NRIS
J1399	Solar	90	METC	Delhi - Churchill - Tompkins 138kV Line	Ingham County	MI	NRIS
J1401	Solar	150	METC	Batavia - Wagner 138kV Line	Branch County	MI	NRIS
J1403	Solar	200	METC	Marquette 138 kV Substation	Ionia County	MI	NRIS
J1430	Solar	100	METC	Verona - Foundary 138kV Line	Calhoun County	MI	NRIS
J1472	Solar	100	METC	Leoni - Parr Raod 138kV line	Jackson County	MI	NRIS



1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2019 Cycle2 Michigan Phase 2 reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in [Appendix A – Cost Allocation Summary \(CEII\)](#)

Table 2: Total Cost of Network Upgrades for DPP 2019 Cycle2 Michigan Phase 2 Projects

Project	DPP Phase II									Payment in DPP Phase I		Payment in DPP Phase II		Interconnection Facilities - Self Fund
	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Cost of NU (\$)	M2 (\$)	M3 =10% of NU-M2 (\$)	20% of Total Cost of NU (\$)	M4= 20% of NU-M2-M3 (\$)	
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO-Owned Direct Assigned							
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i					
J1192						\$30,168.00	\$0.00		\$30,168.00	\$140,000.00	\$0.00	\$6,033.60	\$0.00	NA
J1194						\$2,000,250.00	\$299,214.00		\$2,000,250.00	\$900,800.00	\$0.00	\$400,050.00	\$0.00	No
J1196						\$12,620,784.00	\$685,764.00		\$12,620,784.00	\$800,000.00	\$230,000.00	\$2,524,156.80	\$1,494,156.80	No
J1203						\$4,245,703.00	\$651,516.00		\$4,245,703.00	\$500,000.00	\$0.00	\$849,140.60	\$349,140.60	No
J1207						\$1,413,096.00	\$346,620.00		\$1,413,096.00	\$200,000.00	\$0.00	\$282,619.20	\$82,619.20	No
J1210						\$11,049,936.00	\$220,008.00		\$11,049,936.00	\$200,000.00	\$480,000.00	\$2,209,987.20	\$1,529,987.20	No
J1224						\$6,578,197.00	\$297,264.00		\$6,578,197.00	\$680,000.00	\$0.00	\$1,315,639.40	\$635,639.40	No
J1226				\$759,506.95		\$4,260,888.00	\$507,744.00		\$4,260,888.00	\$800,000.00	\$0.00	\$852,177.60	\$52,177.60	No
J1248	\$2,371,105.55					\$9,666,463.00	\$359,784.00		\$12,037,568.55	\$400,000.00	\$310,779.97	\$2,407,513.71	\$1,696,733.74	No
J1255					\$3,447,170.99	\$14,427,446.00	\$296,854.00		\$17,874,616.99	\$800,000.00	\$201,666.67	\$3,574,923.40	\$2,573,256.73	No
J1262						\$0.00	\$0.00		\$0.00	\$100,000.00	\$0.00	\$0.00	\$0.00	NA
J1292					\$5,231,789.05	\$27,162,724.00	\$296,854.00		\$32,394,513.05	\$789,600.00	\$367,066.67	\$6,478,902.61	\$5,322,235.94	No
J1297				\$150,000.00		\$18,089,069.00	\$268,272.00		\$18,089,069.00	\$680,000.00	\$0.00	\$3,617,813.80	\$2,937,813.80	NA
J1307					\$324,415.17	\$13,529,222.00	\$773,136.00		\$13,853,637.17	\$600,000.00	\$430,000.00	\$2,770,727.43	\$1,740,727.43	No
J1310	\$2,341,143.07				\$225,188.62	\$14,648,128.00	\$273,059.00		\$17,214,459.69	\$500,000.00	\$1,260,411.30	\$3,442,891.94	\$1,682,480.64	No
J1319						\$0.00	\$0.00		\$0.00	\$487,200.00	\$0.00	\$0.00	\$0.00	NA
J1320	\$3,400,000.00					\$9,261,533.00	\$158,040.00		\$12,661,533.00	\$600,000.00	\$84,177.14	\$2,532,306.60	\$1,848,129.46	No
J1329						\$6,578,197.00	\$297,264.00		\$6,578,197.00	\$80,000.00	\$435,000.00	\$1,315,639.40	\$800,639.40	No
J1331						\$2,881,189.00	\$206,484.00		\$2,881,189.00	\$400,000.00	\$20,000.00	\$576,237.80	\$156,237.80	No
J1350				\$205,726.69		\$13,352,970.00	\$782,602.00		\$13,352,970.00	\$600,000.00	\$430,000.00	\$2,670,594.00	\$1,640,594.00	No
J1375				\$200,000.00		\$14,957,707.00	\$206,484.00		\$14,957,707.00	\$400,000.00	\$350,000.00	\$2,991,541.40	\$2,241,541.40	No
J1379				\$325,000.00	\$4,159,570.72	\$50,088,736.00	\$202,452.00		\$54,248,306.72	\$800,000.00	\$710,000.00	\$10,849,661.34	\$9,339,661.34	No
J1389						\$12,375,600.00	\$684,000.00		\$12,375,600.00	\$480,000.00	\$550,000.00	\$2,475,120.00	\$1,445,120.00	No
J1399						\$9,631,564.00	\$295,044.00		\$9,631,564.00	\$360,000.00	\$0.00	\$1,926,312.80	\$1,566,312.80	NA
J1401						\$5,100,000.00	\$500,000.00		\$5,100,000.00	\$600,000.00	\$17,780.00	\$1,020,000.00	\$402,220.00	No
J1403					\$12,554,669.24	\$1,856,843.00	\$255,156.00		\$14,411,512.24	\$800,000.00	\$0.00	\$2,882,302.45	\$2,082,302.45	No
J1430	\$2,387,751.38					\$8,449,201.00	\$199,890.00		\$10,836,952.38	\$400,000.00	\$0.00	\$2,167,390.48	\$1,767,390.48	No
J1472					\$274,811.38	\$7,130,467.00	\$359,784.00		\$7,405,278.38	\$400,000.00	\$1,242,828.38	\$1,481,055.68	\$0.00	No
TOTAL	\$10,500,000.00	\$0.00	\$0.00	\$1,640,233.64	\$26,217,615.17	\$281,386,081.00	\$9,423,289.00	\$0.00	\$318,103,696.17					

Note 1: Section 8.6 of this study report provides details for J1379 potential costs (instead \$31 Million or ~ \$325 Thousand) to address issues seen in Affected System Study that are contingent on METC upgrades.

Note 2: J1379POI - North Belding Circuit cut in (stability upgrade related to a short circuit ratio at the POI) which was designed as a part of J1379 PGIA study has been included in the NRIS and has been tested in this study for limited contingencies. This will be tested fully in Phase 3. As it was designed very late in Phase 2, it was tested in only NRIS to analyze the impact of J1379 third outlet on mitigations.

Note 3: Section 6.3 of this study report provides details for J1472 potential costs (~ \$2.8 Million) to address issues seen in Stability study. Please refer to Appendix B.



Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 3. No Shared Network Upgrades was identified for this cycle.

Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Element Name	Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)	A10/ISD
Batavia - J1401 138kV	MTEP 17 Project 11884 in service 12/31/2021 raises limit to 288 MVA SN, 332 MVA SE	METC	J1207,J1401		\$0.00	12/31/2021
Benston Junction - Peterson Junction 138kV	Raise the sag limit of the 336 ACSR to 110 MVA- adjusted to 110 MVA (from 111MVA in Phase 1) based on Phase 2 results	METC	J1375		\$0.00	
Bingham - Marquette 138kV (Hubbardston Road Junction - David Junction)	Rebuild about 1.3 miles of 336 ACSR to 954 ACSR. (288MVA, 332 MVA SE)	METC		J1255,J1292,J1379,J1403	\$1,742,000.00	
Bingham - Marquette 138kV (Hubbardston Road Junction - Technical Drive Junction)	Rebuild about 15.2 miles of 336 ACSR to 954 ACSR (288 MVA SN, 332 MVA SE)	METC		J1255,J1292,J1379,J1403	\$20,368,000.00	
Bingham-Marquette (Bingham-Techdr J) 138 kV	Rebuild the 1.8 miles of 336 ACSR with 954 ACSR (new limit 249MVA SN, 249 MVA SE)	METC		J1255,J1292,J1403	\$3,283,200.00	
Blackstone - Stable 138kV (Blastone - Maines Road Junction)	Fix terminal equipment at Blackstone 138kV (new limit 288MVA SN, 332SE)	METC	J1310		\$0.00	
Bullock - Shale 138kV (Shale - Laundra Junction)	Incorrect Rating, Shale project has a rating of 236 MVA, 236 MVA SE	METC		J1194	\$0.00	
Canif - Stephens 345kV	Can be controlled by operating phase shifter	ITCT		J1196,J1319,J1331	\$0.00	
Caniff 120kV phase shifter	Can be controlled by operating phase shifter	ITCT		J1196,J1203,J1319,J1331	\$0.00	
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE - Still needed NRIS	METC		J1248,J1310,J1399,J1472	\$0.00	12/31/2022
Cobb - Peterson Junction 138kV	Raise the sag limit of the 336 ACSR to 110 MVA - Still needed ERIS, NRIS adjusted to 110 MVA (from 111MVA in Phase 1) based on Phase 2 results	METC		J1375	\$0.00	
Coldwater - Ash Road Junction 138kV	MTEP 20 project 18084 Raises sag to 166 MVA	METC	J1320		\$0.00	12/31/2023
Coldwater - J1320 138kV	Rebuild 2 miles of 336 and 477 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1320		\$3,400,000.00	
Eureka - Derby Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1379		\$0.00	6/1/2025
J1379 - Deja Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1379,J1403		\$0.00	6/1/2025
J1379 - Derby Junction 138kV	MTEP 21 Target A Project 20016 Rebuilds the 336 to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1379		\$0.00	6/1/2025
J1401 - Wagner 138kV	MTEP 17 Project 11884 in service 5/14/2021 raises limit to 288 MVA SN, 332 MVA SE	METC	J1320,J1401		\$0.00	12/31/2021
J1472 - Leoni 138kV	Raise the sag limits of the 795 ACSR and the 636 ACSR to 194 MVA. - Still needed NRIS	METC		J1472	\$0.00	
Lark - Wasthenaw Junction 138kV	Raise sag limit to 172 MVA - limit lowered to 172 MVA (195 MVA phase 1) based on phase 2 ERIS results	METC	J1310,J1472		\$0.00	
Leoni - Lark 138kV (Leoni - Washtenaw Junction 138kV)	Raise the Sag limit of the 477 ACSR to 171 MVA, 179MVA SE - Changed from rebuild in Phase 1, Sag remediation needed for NRIS. Fixing 2 POIs raise the limit to (181 MVA	METC		J1310,J1472	\$500,000.00	
Moore Road - Ash Road Junction 138kV	MTEP 20 project 18084 Raises sag to 166 MVA	METC	J1320		\$0.00	12/31/2023
Tangent - Whiting (Whiting - Raisin J) 138 kV	Incorrect Rating, 126 MVA, 126 MVA SE	METC		J1472	\$0.00	
Verona - Mud Lake #1 138kV	Rebuild about 4.4 miles of 336 ACSR to 954 ACSR. (288 MVA SN, 332 MVA SE)	METC	J1248,J1310,J1430		\$7,100,000.00	
White Lake - Cobb 138kV (Cobb - Peterson Junction)	Rating should be 110 MVA	METC	J1375		\$0.00	

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) Analysis performed shows there are no study projects for Shared Network Upgrade cost allocation.
- 4) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, stated that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to J1192, J1194, J1196, J1203, J1207, J1210, J1224, J1226, J1248, J1255, J1262, J1292, J1297, J1307, J1310, J1319, J1320, J1329, J1331, J1350, J1375, J1379, J1389, J1399, J1401, J1403, J1430, and J1472 projects included in the DPP 2019 Cycle2 Michigan study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading power factor, however, four projects do not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in Table 4. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 4: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1192	35	+/-11.504	1 X 7	34.6	17.48	0.89	34.47	-12.26	-0.94	Yes	N/A
J1194	225.2	+/-109.07	#N/A	221.05	80.3	0.94	220.47	-73.72	-0.95	Yes	N/A
J1196	200	+/-108.00	#N/A	200.03	70.7	0.94	198.9	-146.26	-0.81	Yes	N/A
J1203	125	+/-45.5	4 X 12	123.36	54.39	0.92	122.99	-60.73	-0.90	Yes	N/A
J1207	50	+/-37.495	#N/A	50.07	37.49	0.80	50.07	-18.75	-0.94	Yes	N/A
J1210	50	+/-87.815	1 X 10	49.72	97.72	0.45	49.29	-53	-0.68	Yes	N/A
J1224	170	+/-82.335	1 X 5	188.27	62.87	0.95	187.93	-92.63	-0.90	Yes	N/A
J1226	200	+/-150	2 X 13	98.6	110.2	0.67	98.36	-40.58	-0.92	Yes	N/A
J1248	100	+/-78.75	#N/A	100.69	52.22	0.89	100.69	-52.17	-0.89	Yes	N/A
J1255	100	+/-65.736	1 X 3	99.5	63.19	0.84	99.54	-79	-0.78	Yes	N/A
J1262	25	+/-10	#N/A	430.33	203	0.90	436	-428	-0.71	Yes	N/A
J1292	198	+/-13.2911	#N/A	196.5	70.19	0.94	194.6	115.4	-0.86	Yes	N/A
J1297	170	+/-96.343	#N/A	145.1	56.23	0.93	142.63	113.32	-0.78	Yes	N/A
J1307	150	+/-85.009	#N/A	148.14	74.51	0.89	147.64	-56.56	-0.93	Yes	N/A
J1310	125	+/-41.09	1 X 42	123.3	71.52	0.87	122.94	-44.21	-0.94	Yes	N/A
J1319	117.5	+/-61.045	1 X 2	118.45	46.2	0.93	114.9	-99.9	-0.75	Yes	N/A
J1320	150	+/-61.616	#N/A	148.54	59.61	0.93	147.22	-73.29	-0.90	Yes	N/A
J1329	20	+/-9.686	#N/A	19.89	8.88	0.91	19.83	-10.98	-0.87	Yes	N/A

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f. (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1331	100	+/-32.87	1 X 28	99.49	65.68	0.83	99.34	-36.7	-0.94	Yes	N/A
J1350	150	+/-61.616	1 X 28	148.54	93.83	0.85	147.79	-69.39	-0.91	Yes	N/A
J1375	100	+/-75	2 X 10	98.82	107.82	0.68	98.64	-38.47	-0.93	Yes	N/A
J1379	200	+/-94.014	#N/A	197.56	83.5	0.92	195.8	-112.57	-0.87	Yes	N/A
J1389	120	+/-61.539	#N/A	118.19	47	0.93	118.05	-86	-0.81	Yes	N/A
J1399	90	+/-51.005	#N/A	88.94	44.82	0.89	88	-63.63	-0.81	Yes	N/A
J1401	150	+/-85.009	#N/A	148.25	75.32	0.89	147.31	-58.81	-0.93	Yes	N/A
J1403	200	+/-110.426	#N/A	98.88	50.08	0.89	98.72	-32.7	-0.95	Yes	N/A
J1430	100	+/-32.868	2 X 7	99.25	44.71	0.91	98.95	-39.36	-0.93	Yes	N/A
J1472	100	+/-32.868	2 X 7	47.95	25.18	0.89	48.46	-25.78	-0.88	Yes	N/A

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2019 Cycle1 Michigan models is the MTEP 19 models with the Bench Cases including all pre-queued projects and associated network upgrades, while the Study Cases contain all of the interconnection requests in DPP 2019 Cycle2 Michigan Phase 2, in addition to all the facilities in the Bench Cases. Table 5 provides the list of different Scenarios study for DPP 2019 Cycle2 Michigan.

- Bench Cases
 - DPP19C2-Phase2-2024SH-Bench
 - DPP19C2-Phase2-2024SUM-Bench
- Study Cases
 - DPP19C2-Phase2-2024SH-Study
 - DPP19C2-Phase2-2024SUM-Study

Table 5: DPP 2019 Cycle1 Michigan-Scenarios

Case Category	Case Name	Case Season	Flow on ITCT-IESO interface	Flow on METC-ATC DC line	Status of Ludington	Load Level
1	DPP19C2-Phase2-2024SUM-Study	Summer Peak 2024	Zero Flow	5 MW ATC to METC	Generating (6 Gen)	Peak
2	DPP19C2-Phase2-2024SH-Study	Summer Shoulder 2024	Zero Flow	5 MW ATC to METC	Generating (5 Gen)	MTEP 18 Base SH (77-78%)

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2019 Cycle2 Michigan generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 34.6.1 and TARA version 2001_2_0_0.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2024 Summer Peak and Summer Shoulder scenarios show generator projects J1248, J1310, J1320, and J1430 causing constraints. The details pertaining to the thermal analysis can be found in Appendix A – Cost Allocation Summary (CEII).



4.1. J1248

The thermal analysis identified the generator to contribute to the following constraint:

1. J758 (Mud Lake) - Verona #1 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$7.1 million, with the project allocated 33.40% of the cost resulting \$2,371,105.55.

4.2. J1310

The thermal analysis identified the generator to contribute to the following constraint:

1. J758 (Mud Lake) - Verona #1 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$7.1 million, with the project allocated 32.97% of the cost resulting \$2,341,143.07.

4.3. J1320

The thermal analysis identified the generator to contribute to the following constraint:

1. J1320 substation to Coldwater 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$3.4 million, with the project allocated 100% of the cost resulting \$3,400,000.00.

4.4. J1430

1. J758 (Mud Lake) - Verona #1 138kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the above listed constraint. The planned upgrade is to rebuild the line. The total cost estimate is \$7.1 million, with the project allocated 33.63% of the cost resulting \$2,387,751.38.

5. Voltage Analysis

The voltage analysis results for 2024 Summer Peak and Summer Shoulder scenarios show that the study generators do not cause any voltage violations. The details pertaining to the voltage analysis can be found in Appendix C – MISO ERIS Analysis (CEII).

6. Stability Analysis

MISO's stability analysis has observed stability issues with J1292, J1379, J1472. Issues seen in J1292, J1379 have been addressed while J1472 needs further data investigation.

6.1. J1292

Non – convergence issues were seen in certain contingencies where J1292 is being isolated to 46 kV Thompson Road. Reconfiguration of line terminals at POI station resolved the issue. The cost of reconfiguration is \$0.0.

6.2. J1379

The ITCT and METC systems were found to be unstable for two fault scenarios simulated titled 3PLO J1379-Vestaburg: Eureka-NBelding Near Eureka and 3PLO J1379-Vestaburg: Eureka-NBelding Near NBelding. However, during the J1379 PGIA Study & Interconnection Facility study, it was identified to cut-in the existing Eureka - Vestaburg 138kV Circuit into the New J1379 Interconnection Station by extending each line into the new station to address protection issues which would also potentially resolve the stability issues. The mitigating solution designed during Phase 2 will be included in the base case models during Phase 3 and will be tested thoroughly.

6.3. J1472

Oscillation issues observed in J1472 are under investigation. Additionally, non-convergence issues were seen in contingencies where J1472 was being islanded with Parr Rd 46kV. If model tuning does not work, then transmission fix would likely require an additional outlet in area. Mitigating solution will be developed and tested in Phase3. Approximate cost would be \$2.8 million.

7. Short Circuit Analysis

Short circuit analysis was performed by ITC utilizing CAPE software. Faults were simulated on the ITC Transmission and METC systems and the results were analyzed to determine the maximum fault current system interrupting devices would be required to clear. Short Circuit study of DPP 2019 April Phase 2 indicates that no project from this cycle adversely impacts the circuit breaker capability. The details pertaining to the short circuit analysis are presented in Appendix E – ITC Short Circuit Analysis (CEII).

8. Affected System Impact Study

The details pertaining to the PJM, CE, DTE, and ATC Affected Systems studies are in, Appendix F – PJM Affected Systems Study Report (CEII), Appendix G – CE Affected Systems Study Report (CEII), Appendix H – DTE Affected Systems Study Report (CEII), and Appendix I – WPSC Affected Systems Study Report (CEII).

8.1. J1226

Per PJM analysis, this generator was found responsible for the following Network Upgrade:

Reconductor the E Frankfort -Crete 345 kV line. The cost estimate is \$13.8 million, with the project allocated 1.91% of the cost. Replace 1-345kV Circuit Breaker, station conductor at both terminals of E Frankfort - Crete 345 kV line. The cost estimate is \$5.2 million, with the project allocated 2.11% of the cost. Replace 2- 345kV switches and 2-345kV circuit breakers and all associated equipment. Relay scheme updated. The cost estimate is \$7.5 million, with the project allocated 3.83% of the cost. Reconductor the line (17STJOHN 345 kV - CRETE EC ;BP 345 kV 1). The cost estimate is \$14.9million, with the project allocated 0.84% of the cost. Add a 2nd Benton 345/138 kV transformer. The cost estimate is \$7.912565 million, with the project allocated 3.0% of the cost.

8.2. J1297

Per CE analysis, this generator was found responsible for the following Network Upgrade:

1. METC Interconnection Protection Requirement

Per CE cost allocation rules, the project receives cost allocation for this upgrade. The planned upgrade is to Modify the existing direct transfer trip (DTT) protection system at the Bell Road Substation to accept additional DTT inputs from the new METC J1297 Switching Station to trip existing. The cost estimate is \$0.15 million, with the project allocated 100% of the cost.

8.3. J1307



Per PJM analysis, this generator was found responsible for the following Network Upgrade: Add a 2nd Benton 345/138 kV transformer. The cost estimate is \$7.912565 million, with the project allocated 4.1% of the cost

8.4. J1375

Per CE analysis, this generator was found responsible for the following Network Upgrade:

1. METC Interconnection Protection Requirement

Per CE cost allocation rules, the project receives cost allocation for this upgrade. The planned upgrade is at White Lake Substation. Install SEL-2505 relays on Transformer Banks #1 & #2 to communicate with new METC 138 kV bus protection; Install aux CT's on Transformer Bank #2. Connect Transformer Bank #2 to METC's new 138 kV Bus #2. The cost estimate is \$0.2 million, with the project allocated 100% of the cost.

8.5. J1350

Per PJM analysis, this generator was found responsible for the following Network Upgrade: Add a 2nd Benton 345/138 kV transformer. The cost estimate is \$7.912565 million, with the project allocated 2.6% of the cost

8.6. J1379

Per CE analysis, this generator is conditional on METC's proposed Network Upgrades to build a new 138 kV circuit from J1379 Switching Station to North Belding Substation.

Per WPSC analysis, this generator was found responsible for the system upgrade required for voltage conversion from Vestaburg to Hersey to 138kV. Per WPSC cost allocation rules, this project receives cost allocation for this upgrade. All transmission facilities on the Vestaburg to Weidman and Weidman to Hersey Breaker-to-breaker line sections will be converted to 138kV operating voltage, including all necessary equipment replacements and line modification. The cost estimate is \$31 million, with the project allocated 100% of the cost. Contingent on METC's proposed Network Upgrades to build a new 138kV circuit from J1379 Switching Station to the North Belding Station, the scope of WPSC required upgrades will be reduced to Vestaburg 69kV Tie Upgrade with an expected cost of \$325 Thousand, with the project allocated 100% of the cost. The total estimated costs for the Wolverine Network Upgrades required to accommodate the J1379 project is \$31,000,000. Consistent with Study Results section, because Wolverine has been informed that the J1379 developer has entered into an Engineering and Procurement with METC to construct a project from Deja to North Belding, Wolverine's upgrades are reduced to the following upgrade at the Vestaburg Transmission Station facility with a cost estimate of \$325,000.00. The cost estimates are based on Good Utility Practice and are in 2021 dollars.

9. Deliverability Analysis

9.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource (ER) project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

9.2. Determining the MW Restriction



If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

9.3. Deliverability Study Results

The limiting constraints (mon-con pairs) seen in the deliverability analysis for the 2024 Summer case are summarized in Appendix J – MISO Deliverability Analysis (CEII)

9.4. J1192

This generator is determined to be fully deliverable for 35 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 6: NRIS Results for J1192 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 6: NRIS Results for J1192

J1192 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					35 MW(100%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.5. J1194

This generator is determined to be fully deliverable for 225.2 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 7 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J1194

J1194 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					225.2 MW(100%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.6. J1196

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J1196

J1196 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					0 MW(0%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)

Canif - Stephens 345kV	Can be controlled by operating phase shifter		200	0.0617			J1196,J1319,J1331	\$0.00	\$0.00
Canif 120 kV phase shifter	Can be controlled by operating phase shifter		200	0.0655			J1196,J1203,J1319,J1331	\$0.00	\$0.00

9.7. J1203

This generator is determined to be fully deliverable for 125 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 9: NRIS Results for J1203

J1203 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)						0 MW(0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Canif 120 kV phase shifter	Can be controlled by operating phase shifter		125	0.0524			J1196,J1203,J1319,J1331	\$0.00	\$0.00

9.8. J1207

This generator is determined to be fully deliverable for 49.5 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 10: NRIS Results for J1207

J1207 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)						49.5 MW(100%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.9. J1210

This generator is determined to be fully deliverable for 50 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 11: NRIS Results for J1210

J1210 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)						50 MW(100%)			
Next Upgrade for Higher NRIS Level (cumulative)	Mitigation /Upgrade	ISD	Level of Service	Distribution Factor	Constraint in ERS Analysis?	Projects Associated	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)

(i.e. All upgrades must be made for 100% NRIS)			Attainable (MW)			with ERIS Constraint			
								\$0.00	\$0.00

9.10. J1224

This generator is determined to be fully deliverable for 170 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 12 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 12: NRIS Results for J1224

J1224 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						170 MW(100%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.11. J1226

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 13 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 13: NRIS Results for J1226

J1226 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						200 MW(100%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.12. J1248

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 14 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 14: NRIS Results for J1248

J1248 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)							6.64 MW(6.64%)				
Next Upgrade for Higher NRIS Level (cumulative)	Mitigation /Upgrade			ISD	Level of Service	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated	Projects Associated with NRIS Constraint	Upgrade Cost Allocated	Total Cost of

(i.e. All upgrades must be made for 100% NRIS)			Attainable (MW)			with ERI Constraint		to Project	Upgrade (\$)
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE - Still needed NRIS	12/31/2022	100	0.0577			J1248,J1310,J1399,J1472	\$0.00	\$0.00

9.13. J1255

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis.

Table 15 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 15: NRIS Results for J1255

J1255 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)						0 MW(0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERI Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Bingham - Marquette 138kV (Hubbardston Road Junction - David Junction)	Rebuild about 1.3 miles of 336 ACSR to 954 ACSR. (288MVA, 332 MVA SE)		52.91	0.0874			J1255,J1292,J1379,J1403	\$229,327.91	\$1,742,000.00
Bingham-Marquette (Bingham-Techdr J) 138 kV	Rebuild the 1.8 miles of 336 ACSR with 954 ACSR (new limit 249MVA SN, 249 MVA SE)		61.82	0.0875			J1255,J1292,,J1403	\$536,470.59	\$3,283,200.00
Bingham - Marquette 138kV (Hubbardston Road Junction - Technical Drive Junction)	Rebuild about 15.2 miles of 336 ACSR to 954 ACSR (288 MVA SN, 332 MVA SE)		100	0.0874			J1255,J1292,J1379,J1403	\$2,681,372.50	\$20,368,000.00

9.14. J1292

This generator is determined to be fully deliverable for 198 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 16 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 16: NRIS Results for J1292

J1292 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)						62.34 MW(31.48%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERI Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Bingham - Marquette 138kV (Hubbardston Road Junction - David Junction)	Rebuild about 1.3 miles of 336 ACSR to 954 ACSR. (288MVA, 332 MVA SE)		148.11	0.0669			J1255,J1292,J1379,J1403	\$347,665.31	\$1,742,000.00
Bingham - Marquette 138kV (Hubbardston Road Junction - Technical Drive Junction)	Rebuild about 15.2 miles of 336 ACSR to 954 ACSR (288 MVA SN, 332 MVA SE)		159.77	0.0669			J1255,J1292,J1379,J1403	\$4,065,009.79	\$20,368,000.00
Bingham-Marquette (Bingham-Techdr J) 138 kV	Rebuild the 1.8 miles of 336 ACSR with 954 ACSR (new limit 249MVA SN, 249 MVA SE)		198	0.0675			J1255,J1292,J1379,J1403	\$819,113.95	\$3,283,200.00

9.15. J1307

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis.

Table 17 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 17: NRIS Results for J1307

J1307 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)						150 MW(100%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERI Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.16. J1310

This generator is determined to be fully deliverable for 125 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 18 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 18: NRIS Results for J1310

J1310 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)							72.46 MW(57.97%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERI Analysis?	Projects Associated with ERI Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE - Still needed NRIS	12/31/2022	98.49	0.1025			J1248,J1310,J1399,J1472	\$0.00	\$0.00	
Leoni - Lark 138kV (Leoni - Washtenaw Junction 138kV)	Raise the Sag limit of the 477 ACSR to 171 MVA, 179MVA SE - Changed from rebuild in Phase 1, Sag remediation needed for NRIS. Fixing 2 POIs raise the limit to (181 MVA	6/1/2025	125	0.0621			J1310,J1472	\$225,188.62	\$500,000.00	

9.17. J1319

This generator is determined to be fully deliverable for 117.5 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 19 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 19: NRIS Results for J1319

J1319 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERI and case assumptions)						0 MW(0%)			
Next Upgrade for Higher NRIS Level			Level of Service	Distribution Factor		Projects Associated	Projects Associated with NRIS Constraint		Total Cost of Upgrade (\$)

(cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Attainable (MW)		Constraint in ERIS Analysis?	with ERIS Constraint		Upgrade Cost Allocated to Project	
Canif - Stephens 345kV	Can be controlled by operating phase shifter		117.5	0.0569			J1196,J1319,J1331	\$0.00	\$0.00
Canif 120 kV phase shifter	Can be controlled by operating phase shifter		117.5	0.0604			J1196,J1203,J1319,J1331	\$0.00	\$0.00

9.18. J1320

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 20 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 20: NRIS Results for J1320

J1320 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					150 MW(100%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.19. J1329

This generator is determined to be fully deliverable for 20 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 21 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 21: NRIS Results for J1329

J1329 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					20 MW(100%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.20. J1331

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 22 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 22: NRIS Results for J1331

J1331 Deliverable (NRIS) Amount in 2024 Case:	0 MW(0%)
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(Conditional on ERS and case assumptions)									
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Canif - Stephens 345kV	Can be controlled by operating phase shifter		100	0.065			J1196,J1319,J1331	\$0.00	\$0.00
Canif 120 kV phase shifter	Can be controlled by operating phase shifter		100	0.0689			J1196,J1203,J1319,J1331	\$0.00	\$0.00

9.21. J1350

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 23 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 23: NRIS Results for J1350

J1350 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)						150 MW(100%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.22. J1375

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 24 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 24: NRIS Results for J1375

J1375 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERS and case assumptions)						100 MW(100%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.23. J1379

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 25 shows the NRIS results and cost estimates determined in the

Table 25: NRIS Results for J1379

J1379 ¹ Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						54.64 MW(27.32%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Bingham - Marquette 138kV (Hubbardston Road Junction - David Junction)	Rebuild about 1.3 miles of 336 ACSR to 954 ACSR. (288MVA, 332 MVA SE)		132.76	0.0624			J1255,J1292,J1379,J1403	\$327,723.75	\$1,742,000.00
Bingham - Marquette 138kV (Hubbardston Road Junction - Technical Drive Junction)	Rebuild about 15.2 miles of 336 ACSR to 954 ACSR (288 MVA SN, 332 MVA SE)		200	0.0624			J1255,J1292,J1379,J1403	\$3,831,846.96	\$20,368,000.00

9.24. J1389

This generator is determined to be fully deliverable for 120 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 26 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 26: NRIS Results for J1389

J1389 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						120 MW(100%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.25. J1399

This generator is determined to be fully deliverable for 90 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 27 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 27: NRIS Results for J1399

J1399 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						17.03 MW(18.92%)			
Next Upgrade for Higher NRIS Level (cumulative)	Mitigation /Upgrade	ISD	Level of Service	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated	Projects Associated with NRIS Constraint	Upgrade Cost Allocated	Total Cost of

¹ J1379POI - North Belding Circuit cut in (stability upgrade related to a short circuit ratio at the POI) which was designed as a part of J1379 PGIA study has been included in the NRIS and has been tested in this study for limited contingencies. This will be tested fully in Phase 3. As it was designed very late in Phase 2, it was tested in only NRIS to analyze the impact of J1379 third outlet on mitigations.

(i.e. All upgrades must be made for 100% NRIS)			Attainable (MW)			with ERIS Constraint		to Project	Upgrade (\$)
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE - Still needed NRIS	12/31/2022	90	0.0738			J1248,J1310,J1399,J1472	\$0.00	\$0.00

9.26. J1401

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 28 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 28: NRIS Results for J1401

J1401 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					150 MW(100%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.27. J1403

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 29 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 29: NRIS Results for J1403

J1403 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)							143.11 MW(71.56%)		
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
Bingham - Marquette 138kV (Hubbardston Road Junction - David Junction)	Rebuild about 1.3 miles of 336 ACSR to 954 ACSR. (288MVA, 332 MVA SE)		173.8	0.1596			J1255,J1292,J1379,J1403	\$837,283.02	\$1,742,000.00
Bingham-Marquette (Bingham-Techdr J) 138 kV	Rebuild the 1.8 miles of 336 ACSR with 954 ACSR (new limit 249MVA SN, 249 MVA SE)		179.08	0.1572			J1255,J1292,J1379,J1403	\$1,927,615.46	\$3,283,200.00
Bingham - Marquette 138kV (Hubbardston Road Junction - Technical Drive Junction)	Rebuild about 15.2 miles of 336 ACSR to 954 ACSR (288 MVA SN, 332 MVA SE)		200	0.1596			J1255,J1292,J1379,J1403	\$9,789,770.75	\$20,368,000.00

9.28. J1430

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 30 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 30: NRIS Results for J1430

J1430 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)					100 MW(100%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
								\$0.00	\$0.00

9.29. J1472

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 31 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 31: NRIS Results for J1472

J1472 Deliverable (NRIS) Amount in 2024 Case: (Conditional on ERIS and case assumptions)						82.61 MW(82.61%)				
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mitigation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	
Leoni - Lark 138kV (Leoni - Washtenaw Junction 138kV)	Raise the Sag limit of the 477 ACSR to 171 MVA, 179MVA SE - Changed from rebuild in Phase 1, Sag remediation needed for NRIS. Fixing 2 POIs raise the limit to (181 MVA	6/1/2025	87.95	0.0947			J1310,J1472	\$274,811.38	\$500,000.00	
Cement City Junction - Seamless 138kV	MTEP 20 15939 increases the limit to 227 MVA SN, 286 MVA SE - Still needed NRIS	12/31/2022	96.63	0.1821			J1248,J1310,J1399,J1472	\$0.00	\$0.00	
J1472 - Leoni 138kV	Raise the sag limits of the 795 ACSR and the 636 ACSR to 194 MVA. - Still needed NRIS		100	0.9535			J1472	\$0.00	\$0.00	

10. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis tests for Network Upgrades driven by higher queued interconnection projects was performed for this System Impact Study. No SNU was identified for this cycle.

11. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

11.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

11.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Post Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Project A cost portion of NU} \\ \text{Cost of NU} = \frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.

12. Revisions to the Draft report

1. Rev.1
 - a. Table 2 – M3 & M4 details have been updated. M3 details were updated to match Phase1 report of this study cycle which further impacted M4.
2. Rev. 2
 - a. Table 2 – Added AFS costs (CE & WPSC) – which further impacted total costs of NU and M4.
 - b. Updated Appendix A – Cost Allocation Summary (CEII) phase 2 (Rev2)
 - c. Updated Appendix B – Network Upgrade Table Per Project (CEII)_phase2 (Rev1)
3. Rev. 3
 - a. In this version, NRIS analysis was re-run with ERIS mitigations as a starting case and then any further constraints seen were addressed by NRIS mitigating solutions. In the re-run analysis, ERIS mitigation cost allocation is being done by considering highest MW impact of ERIS projects only. And NRIS mitigation cost allocation is done by considering NRIS highest MW impact of the corresponding study project(s). In the earlier version, NRIS analysis was run prior to applying the ERIS mitigation and each constraint/study project and associated DFAX/MW impact was noted. For cost allocation of a mitigating solution, the highest MW impact from either ERIS or NRIS study was considered. The revised cost along with notes is summarized in Table 2. Appendices A, B, J and K provides the details.
 - b. J1379POI - North Belding Circuit cut in (stability upgrade related to a short circuit ratio at the POI) which was designed as a part of J1379 PGIA study has been included in the NRIS and has been tested in this study for limited contingencies. This will be tested fully in Phase 3. As it was designed very late in Phase 2, it was tested in only NRIS to analyze the impact of J1379 third outlet on mitigations. This has impacted the constraints seen in NRIS analysis and is reflected in the study report & Appendices
 - c. Section 8.3 J1379 has been updated with details.
 - d. DTE AFS has been included in Appendix H
4. Rev.4
 - a. Table 2 has been updated to reflect the updated NU and TOIF costs as per latest version of FAS study reports and Appendix B has also been updated to reflect the same.
 - b. The column 'Upgrade cost allocated to project has been updated in Table 15, Table 16, Table 22 and Table 25 to match the cost reflected in Table 2 of this report. Appendix J is also updated to reflect the same.
 - c. Added PJM -AFS details in Table 2 and Section 8 of this report.
5. Rev. 5
 - a. Table 2 has been updated to reflect the PJM AFS cost to J1350 which was incorrectly shown as J1350 NRIS cost.
 - b. Updated J1255, J1292, J1379, J1430 TONU cost in Table 2 and in Appendices A & B
 - c. Updated NRIS costs of J1255, J1292, J1310, J1379, J1403, J1472 and thus updated costs in Table 2 and Appendices A, B, J.
 - d. J1379 AFS costs in Table 2 updated to 325,000



Appendix A – Cost Allocation Summary (CEII)

Appendix B – Network Upgrade Table Per Project (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D- MISO Stability Analysis

Appendix E – ITC Short Circuit Analysis (CEII)

Appendix F – PJM Affected Systems Study Report (CEII)

Appendix G – CE Affected Systems Study Report (CEII)

Appendix H – DTE Affected Systems Study Report (CEII)

Appendix I – WPSC Affected Systems Study Report (CEII)

Appendix J – MISO Deliverability Analysis (CEII)

Appendix K– MISO A10-Conditionality Analysis (CEII)